



SCSM2223- CROSS-PLATFORM APPLICATION DEVELOPMENT

Chapter 8: Individual Lab Assignment Report (Employee Directory)

Name	LEE MAY YAN
Matric Number	A24CS0101
Section	04
Lecturer	DR. HISHAMMUDDIN BIN ASMUNI

1.0 Screenshots of Working UI

1.1 Main Employee Directory Page

Employee Directory

Manage employee records for the HR office.

Total Active Employees
6

Add Employee

Employee ID

Name

Email

Department

Select department

Position

Hire Date

Salary RM

☐ Active employee

Add Employee

Sort by Salary

Ascending

Apply

Employee List

Employee ID	Name	Email	Department	Position	Hire Date	Salary	Status	Actions
EMP007	Nur Izzati	nur.izzati@company.com	HR	Recruitment Officer	2024-01-31	RM 3,200.00	Active	EditDelete
EMP005	Mei Ling	mei.ling@company.com	Marketing	Marketing Executive	2023-04-11	RM 3,900.00	Active	EditDelete

Figure 1.1: Screenshot of Main Employee Directory Page

1.2 Form Validation

Employee Directory

Manage employee records for the HR office.

Total Active Employees
6

Add Employee

Employee ID
E001
Employee ID must be like EMP001.

Name
JENNIE

Email
jennie@gmail
Please enter a valid email.

Department
IT

Position
Software Engineer

Hire Date
23/05/2026
Hire date cannot be in the future.

Salary RM
1500

☒ Active employee

Add Employee

Figure 1.2: Screenshot of Form Validation (Invalid Employee ID, email and hire date)

1.3 CRUD Operation

Edit Employee

Employee ID
EMP001

Name
Aina Rahman Updated

Email
aina.rahman@company.com

Department
HR

Position
Senior HR Executive

Hire Date
14/03/2021

Salary RM
4000.00

☐ Active employee

Update Employee Cancel

Figure 1.3: Screenshot of Edit Employee Operation

1.4 Search and Sort Functionality

Sort by Employee ID

Ascending

Apply

Employee List

Employee ID	Name	Email	Department	Position	Hire Date	Salary	Status	Actions
EMP008	JENNIE	jennie@gmail.com	IT	Software Engineer	2026-04-30	RM 1,500.00	Active	EditDelete
EMP009	John	helioganier@gmail.com	IT	Software Engineer	2026-04-01	RM 4,000.00	Active	EditDelete

Figure 1.4: Screenshot of Search and Sort Functionality

2.0 Achievement of Learning Outcomes

1. LO1 - Connect Vue 3 application to an external REST API

The Vue 3 frontend connects to the Express backend via REST API calls. Employee information will be fetched dynamically and displayed into the UI. Create, update, delete, search and sort operations are all performed using API calls.

2. LO2 - Configure and use the Axios HTTP client

A reusable Axios instance is used within the src/services/api.js file and is not done within components through direct axios function calls. Axios defines the base URL, request timeout and JSON headers as defaults.

3. LO3 - Build Vue forms using v-model, validation and events

Vue 3 Composition API and v-model are used in EmployeeForm.vue component. Inline error messages are displayed in case of invalid input, while the employee form interacts with its parent components via emitted save/cancel event

4. LO4 - Build Express REST API with CRUD, search, sort, and MySQL

The Express app executes all CRUD operations by using MySQL and Prepared Statements. Server-side sorting & search functionality is supported through query parameters.

3.0 Challenges & Solutions

1. Frontend and backend running on different ports:

Vue frontend runs on port 5174, whereas the Express backend works on port 3001. It led to cross-origin communication issues, which were handled using CORS middleware in Express and Axios with the base URL being set at <http://localhost:3001>.

2. MySQL connection and local environment setup:

The backend must successfully connect to the local Laragon MySQL server before CRUD operations can be performed. The parameters used for connecting to the server are host '127.0.0.1', port 3306, user 'root', and database 'employee_directory'. The schema.sql file creates the database, table employees, and populates the database with seed data.

3. Secure server-side sorting functionality:

Sorting is needed for the API but letting clients sort by any random column names would be a major vulnerability. It was managed by checking the received sortBy parameter against a list of valid columns: name, hireDate, and salary.